3.1. Notes

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1. Learning outcomes

By the end of this lesson, you should be able to:

- List and describe the hardware components used in PLC systems

Prescribed Reading

Frank Petruzella, Programmable Logic Controllers.
McGraw-Hill Higher Education, 5th Edition, 2017. Print
ISBN: 9780073373843

Chapter 2, Section 2.1 - 2.11

2. PLC Hardware Components

2.1. I/O Section

The input/output (I/O) section of a PLC is the section to which all field devices are connected and provides the interface between them and the CPU. Input/output arrangements are built into a fixed PLC while modular types use external I/O modules that plug into the PLC.

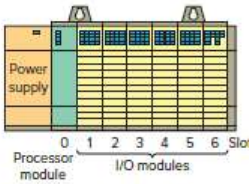

Figure 2-1 Rack-based I/O section.

Each slot in the rack is capable of accommodating any type of I/O module. The I/O system provides an interface between the hardwired components in the field and the CPU. The input interface allows status information regarding processes to be communicated to the CPU and thus allows the CPU to communicate operating signals through the output interface to the process devices under its control.

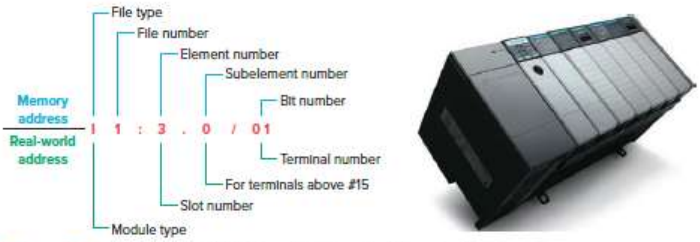

Figure 2-3 Allen-Bradley SLC 500 rack/slot-based addressing format.
Source: Image Courtesy of Rockwell Automation, Inc.

The PLC's memory system stores information about the status of all the inputs and outputs. To keep track of all this information, it uses a system called **addressing**. An address is a label or number that indicates where a certain piece of information is located in a PLC's memory.

Examples of addressing schemes include rack/slot-based, versions of which are used in Allen-Bradley SLC 500 controllers, tag-based used in Allen-Bradley ControlLogix controllers, and PC-based control used in soft PLCs.

In general, rack/slot-based addressing elements include:

- **Type**—The type determines if an input or output is being addressed.

- **Slot—**The slot number is the physical location of the I/O module. This may be a combination of the rack number and the slot number when using expansion racks.
- **Word and Slot—**The word and slot are used to identify the actual terminal connection in a particular I/O module. A discrete module usually uses only one word, and each connection corresponds to a different bit that makes up the word.

2.2. Discrete I/O Modules

The most common type of I/O interface module is the discrete type. This type of interface connects field input devices of the ON/OFF nature such as selector switches, pushbuttons, and limit switches. Likewise, output control is limited to devices such as lights, relays, solenoids, and motor starters that require simple ON/OFF switching.

Figure 2-11 Discrete input and output devices.

Each discrete I/O module is powered by some field-supplied voltage source. Since these voltages can be of different magnitude or type, I/O modules are available at various AC and DC voltage ratings.

Table 2-1 Common Ratings for Discrete I/O Interface Modules

Input Interfaces	Output Interfaces
12 V **AC/DC** /24 V **AC/DC**	12–48 V **AC**
48 V **AC/DC**	120 V **AC**
120 V **AC/DC**	230 V **AC**
230 V **AC/DC**	120 V **DC**
5 V **DC** (TTL level)	230 V **DC**
	5 V **DC** (TTL level)
	24 V **DC**

Discrete output modules are used to turn field output devices either on or off. These modules can be used to control any two-state device, and they are available in AC and DC versions and in various voltage ranges and current ratings. Output modules can be purchased with **transistors, triac, or relay outputs**.

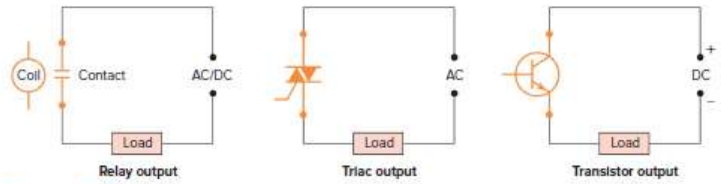

Figure 2-18 Relay, transistor, and triac switching elements.

2.3. Analog I/O Modules and Special I/O Modules

Analog I/O Modules

Discrete devices are inputs and outputs that have only two states: on and off. In comparison, analogue devices represent physical quantities that can have an infinite number of values. Typical analogue inputs and outputs vary from 0 to 20 mA, 4 to 20 mA, or 0 to 10 V.

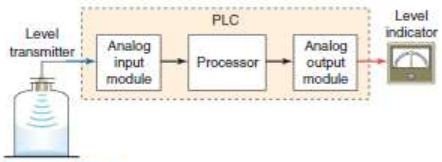

Figure 2-21 Analog input and output to a PLC.

In the typical example above, The analogue input interface module contains the circuitry necessary to accept an analogue voltage or current signal from the level transmitter field device. This input is converted from an analogue to a digital value for use by the processor. The circuitry of the analogue output module accepts the digital value from the processor and converts it back to an analogue signal that drives the field tank level meter.

Special I/O Modules

Many different types of I/O modules have been developed to meet special needs. These include:

- **High-Speed Counter Module** used to provide an interface for applications requiring counter speeds that surpass the capability of the PLC ladder program.

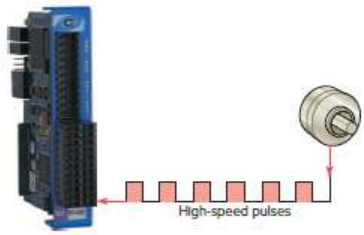

Figure 2-30 High-speed counter module.
Source: Courtesy Control Technology Corporation.

- **Thumbwheel Module** used to allow the use of thumbwheel switches for feeding information to the PLC to be used in the control program.

Figure 2-31 Thumbwheel switch.
Source: Photo courtesy Omron Industrial Automation, **www.ia.omron.com.**

- **TTL Module** allows the transmitting and receiving of TTL (Transistor-Transistor-Logic) signals. This module allows devices that produce TTL-level signals to communicate with the PLC's processor.
- **Encoder-Counter Module** allows the user to read the signal from an encoder on a real-time basis and stores this information so it can be read later by the processor.

Figure 2-32 Encoder.
Source: Photo courtesy of Allied Motion Technologies, Inc.

- **Basic or ASCI Module** runs user-written BASIC and C programs. These programs are independent of the PLC processor and provide an easy, fast interface between remote foreign devices and the PLC processor. Typical applications include interfaces to bar code readers, robots, printers, and displays.
- **Stepper-Motor Module** provides pulse trains to a stepper-motor translator, which enables control of a stepper-motor.

Figure 2-33 Stepper-motor.
Source: Courtesy Sherline Products.

- **BCD-output Module** enables a PLC to operate devices that require BCD-coded signals such as seven- segment displays

Figure 2-34 Seven-segment display.
Source: Courtesy Red Lion Controls.

- **PID Module** is used in process control applications that incorporate PID algorithms.. The basic function of this module is to provide the control action required to maintain a process variable such a temperature, flow, level, or speed within set limits of a specified set point.

Figure 2-35 PID module.
Source: Courtesy Red Lion Controls.

- **Motion and Position Control Module** used in applications involving accurate high-speed machining and packaging operations. Intelligent position and motion control modules permit PLCs to control stepper and servo motors.

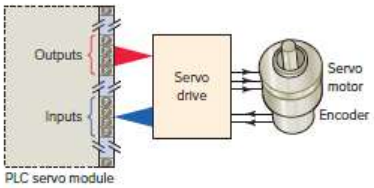

Figure 2-36 PLC servo module.

- **Communication Modules** used to establish point-to-point connections with other intelligent devices for the exchange of data. Such connections are normally established with computers, operator stations, process control systems, and other PLCs.

Figure 2-37 Serial communications module.
Source: Photo courtesy Automation Direct, **www.automationdirect.com.**

2.4. The Central Processing Unit (CPU)

The central processing unit (CPU) is built into single-unit fixed PLCs while modular rack types typically use a plugin module. **CPU, controller**, and **processor** are all terms used by different manufacturers to denote the same module that performs basically the same functions. Processors vary in processing speed and memory options.

A processor module can be divided into two sections: the **CPU section** and the **memory section**.

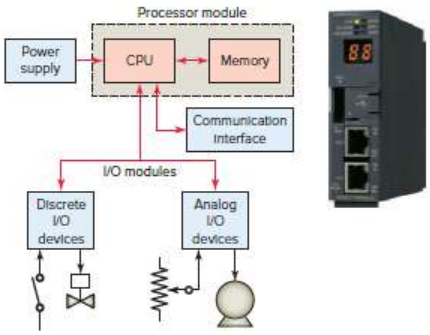

Figure 2-38 Sections of a PLC processor module.
Source: Courtesy Mitsubishi Automation.

The PLC power supply provides the necessary power (typically 5 VDC) to the processor and I/O modules plugged into the backplane of the rack. The CPU contains a similar type of microprocessor found in a personal computer. The difference is that the program used with the microprocessor is designed to facilitate industrial control rather than provide general-

purpose computing.

The Central Processing Unit

- **executes the operating system,**
- **manages memory,**
- **monitors inputs,**
- **evaluates the user logic (ladder program),**
- **and turns on the appropriate outputs.**

The CPU of a PLC system may contain more than one processor. One advantage of using multiprocessing is that the overall operating speed is improved. Each processor has its own memory and programs, which operate simultaneously and independently.

Associated with the processor unit will be a number of status LED indicators to provide system diagnostic information to the operator.

A key switch may be provided that allows you to select one of the following three modes of operation: **RUN**, **PROG**, and **REM**.

RUN mode

It is the normal operation mode of the PLC. In this mode, the ladder program executes and energises the output devices based on the

programmed logic.

- In this mode, online program editing is not possible and prevented.
- The use of a programmer or operator interface device to change the processor mode is disabled.

PROG mode

This is the program mode of the PLC. In this mode, the processor is prevented from scanning or executing the ladder program.

The controller outputs are de-energised.

- The PLC can be reprogrammed, and the ladder logic can thus be modified.
- The user is still unable to use a programmer or operator interface device to change the processor mode.

REM mode

This is the remote mode of the PLC enabling distant control of the device. It can be either be the REMOTE Run mode, REMOTE program or Remote Test Mode.

- The operator is able to change the processor mode from a programmer/operator interface device.
- This mode allows the operator to remotely perform online program editing thus changing the ladder logic.

2.5. Human Machine Interfaces

In the past, the typical user interface to a control system consisted of a panel with switches, pushbuttons, pilot lights, gauges, analog meters, and the like. With the advent of digital control systems, larger hard-wired panels have been replaced by a computer screen with process graphics and operator commands entered via a keyboard.

Figure 2-50 Human Machine Interface (HMI).
Source: Courtesy of Nercon.

Human-machine interfaces give the ability to the operator and to management to view the operation in real-time. Through personal computer–based setup software, you can configure display screens to:

- Replace hardwired pushbuttons and pilot lights with realistic-looking icons. The machine operator need only touch the display panel to activate the pushbuttons.
- Show operations in graphic format for easier viewing.
- Allow the operator to change the timer and counter presets by touching the numeric keypad graphic on the touch screen.
- Show alarms, complete with time of occurrence and location.
- Display variables as they change over time.

Figure 2-53 HMI installed in an industrial environment.
Source: Photo Courtesy PC Enclosures, **http://www.pcenclosures.net.**

HMI operator panels typically contain monochrome or 256-color display screens. These systems often communicate directly with the PLC to read or write memory locations.